FORM HS-05 · REV. 4 POST-STASIS REFRESHER · MODULE 5 OF 9 POST IN EVERY FAB YARD AND WELDING BAY

DUNEWRIGHT COLONY WORKS
Settlement Site 4 · Construction Division · Health & Safety

SPARKS TRAVEL.

WELDING AND CUTTING: THE PERMIT, THE FIRE WATCH, YOUR EYES, YOUR LUNGS AND YOUR CYLINDERS.

NO PERMIT?

NO ARC. NO FLAME. NO EXCEPTIONS.

Any welding, cutting, grinding, brazing or torch work outside a designated welding bay needs a **hot work permit**. The permit is signed by your foreman (white hat) and posted at the work spot. It is good for **one location and one shift**. Brown hats do the work. A red hat stands fire watch.

HOT WORK PERMIT

IN THIS ORDER, EVERY TIME

1. **Look for a way to avoid hot work.** Can the part be bolted, cut cold or moved to a fab yard bay? If yes, do that.
2. **Clear 35 feet.** Move every combustible at least 35 ft away. If it can't move, cover it with fire blankets or metal shields.
3. **Seal openings.** Cover floor gaps, wall holes and deck seams within 35 ft. Sparks fall through and start fires on the deck below.
4. **Check the other side.** Heat moves through walls, pipes and hull plate. Look and clear the far side of whatever you weld.
5. **Test the air** in enclosed modules or near tanks. No hot work if flammable gas is above 10% of the lower explosive limit (LEL).
6. **Post the fire watch.** Red hat, charged extinguisher in hand, on Ch 13, eyes on the sparks. Fire watch does no other job.
7. **Foreman signs and posts** the permit at the spot. Call it in on Ch 13: "Hot work starting, Echo-22, Deck 4."
8. **After the last arc: fire watch stays 60 minutes.** Then a final walk of the area, and the permit is closed out on Ch 13.

Old Earth rule (OSHA 1910.252, NFPA 51B) is a fire watch during work and at least 30 minutes after. DCW site rule is 60 minutes, because dry sand-sealed insulation and polymer panels smolder before they flame.

THE 35-FOOT RULE

35 ft
CLEAR OR COVER

- Sparks and slag fly **more than 35 ft**, farther from a height and in wind.
- Work from Deck 3 or above? Clear the area below too, or catch the slag with a fire blanket.
- Wind over 20 mph or a blowing-dust advisory on Ch 16: **outdoor hot work stops.**

FIRE WATCH DUTIES

- **Red hard hat.** Charged extinguisher in reach. Know where the next one is.
- **Watch the sparks,** not the weld. Look where they land, including below and behind.
- **Stop the work** the moment anything smokes or catches. You have the authority.
- **Fire you can't put out in 10 seconds:** Ch 1, "Emergency, emergency, emergency. Fire," with sector-grid and deck. Get everyone out.
- Stay **60 minutes** after the last arc. Feel walls and panels near the work for heat before you leave.

PICK THE RIGHT LENS

FILTER SHADE NUMBERS · DARKER = HIGHER

PROCESS	SIZE / CURRENT	MINIMUM SHADE	SUGGESTED (COMFORT)
STICK WELDING (SMAW)			
Stick	Under 60 A	7	–
Stick	60–160 A	8	10
Stick	160–250 A	10	12
Stick	250–550 A	11	14
WIRE WELDING (GMAW / FCAW)			
MIG / flux core	Under 60 A	7	–
MIG / flux core	60–160 A	10	11
MIG / flux core	160–250 A	10	12
MIG / flux core	250–500 A	10	14
TIG (GTAW)			
TIG	Under 50 A	8	10
TIG	50–150 A	8	12
TIG	150–500 A	10	14
PLASMA ARC CUTTING			
Plasma cut	Under 300 A	8	9
Plasma cut	300–400 A	9	12
Plasma cut	400–800 A	10	14
TORCH (OXYFUEL)			
Torch cutting	Plate under 1 in	3	3–4
Torch cutting	Plate 1–6 in	4	4–5
Torch cutting	Plate over 6 in	5	5–6
Gas welding	Under ⅛ in	–	4–5
Gas welding	⅛–½ in	–	5–6
Gas welding	Over ½ in	–	6–8

Start darker than you think you need, then go lighter until you can see the weld zone. Never go below the minimum. Source: OSHA 1910.252 / ANSI Z49.1 lens shade guide.

ARC EYE

Ultraviolet from the arc burns the surface of the eye. You feel nothing at first. **Hours later**: gritty, burning, watering eyes, light hurts.

DO: report to the Forward Aid Station. Don't rub. Bystanders get it too, so put up welding screens.

FUMES

Stainless steel gives off hexavalent chromium, a carcinogen. Old Earth limit: 5 µg/m³ over a shift. **Manganese** damages nerves.

DO: keep your head out of the plume. Use the fume extractor every time. Respirator when the permit says so.

METAL FUME FEVER

Welding **galvanized** (zinc-coated) steel. Flu-like chills, fever, metal taste, aches, 4–12 hours after work.

DO: grind the coating off first, ventilate, respirator. Report it. It is a sign your exposure is too high.

GAS CYLINDERS

- **Upright and chained** or on a cart. Valve cap on whenever it isn't hooked up.
- **Oxygen stored at least 20 ft from fuel gas**, or separated by a wall at least 5 ft high with a half-hour fire rating.
- **Acetylene never above 15 psig.** Above that it can decompose and explode on its own.
- **Flashback arrestors** on both hoses at the regulator.
- **Leak test with soapy water.** Never with a flame.
- **Never blow off clothes or dust with oxygen.** Oxygen-soaked cloth burns like a torch from one spark.
- Keep cylinders out of direct sun on the deck. Shade racks only.

SHOCK

- Desert sweat is your enemy here. **Wet skin and damp leathers** let current through you.
- **Change to dry gloves** when they get damp. Carry spares.
- **Insulate yourself from the work:** dry rubber mat, dry boards. Never sit or kneel on wet plate.
- Inspect leads before every shift. Cracked insulation gets tagged out (see HS-07).
- Never wrap leads around your body. Never change electrodes with bare hands.

BURNS & FIRE

STOP. DROP.
ROLL. COOL.

Leathers save you from sparks. Synthetic clothing under them melts into skin. Wear cotton or flame-resistant base layers only.

- **Clothing on fire:** stop, drop and roll. Crew smother it with a fire blanket.
- **Burns:** cool with clean running water for **20 minutes**. No ice, butter or ointment.
- **Remove** rings, watches and gear near the burn before it swells. Leave stuck cloth in place.
- **Large, deep or face burns:** Ch 1, "Emergency, emergency, emergency. Fire," then Ch 14 for the medic.

HEAT STACK: DO

- Welding leathers plus desert heat doubles your heat load. Read HS-03 Heat.
- Schedule welding for the **night shift** whenever you can.
- Treat welding in leathers as **heavy work** on the WBGT table.
- Wear a cooling vest under leathers when the heat level is Danger or higher.

HEAT STACK: DON'T

- Don't weld inside a sun-baked module without testing the air and the temperature.
- Don't skip rest breaks because "the bead's almost done."
- Don't strip your leathers to cool off mid-job. Stop, step out, cool down, then suit back up.

ANYONE CAN STOP HOT WORK

Smoke, smell, sparks landing where they shouldn't, no fire watch, or no permit posted: say **"stop"** and call Ch 13. The welder stops. No penalty, no questions.

WORKER ACKNOWLEDGEMENT

I have read this poster. I know where the nearest extinguisher, fire blanket and eyewash are, and I will not start hot work without a posted permit and fire watch.

NAME	COLONIST ID	CREW / SECTOR	DATE

Adapted from Old Earth standards: OSHA 29 CFR 1910.252 & 1926 Subpart J (welding and cutting), NFPA 51B (fire prevention during hot work), ANSI Z49.1 (safety in welding, cutting and allied processes), OSHA 1910.1026 (hexavalent chromium), AWS safety and health fact sheets.

HS-05 · Dunewright Colony Works · Replace when faded